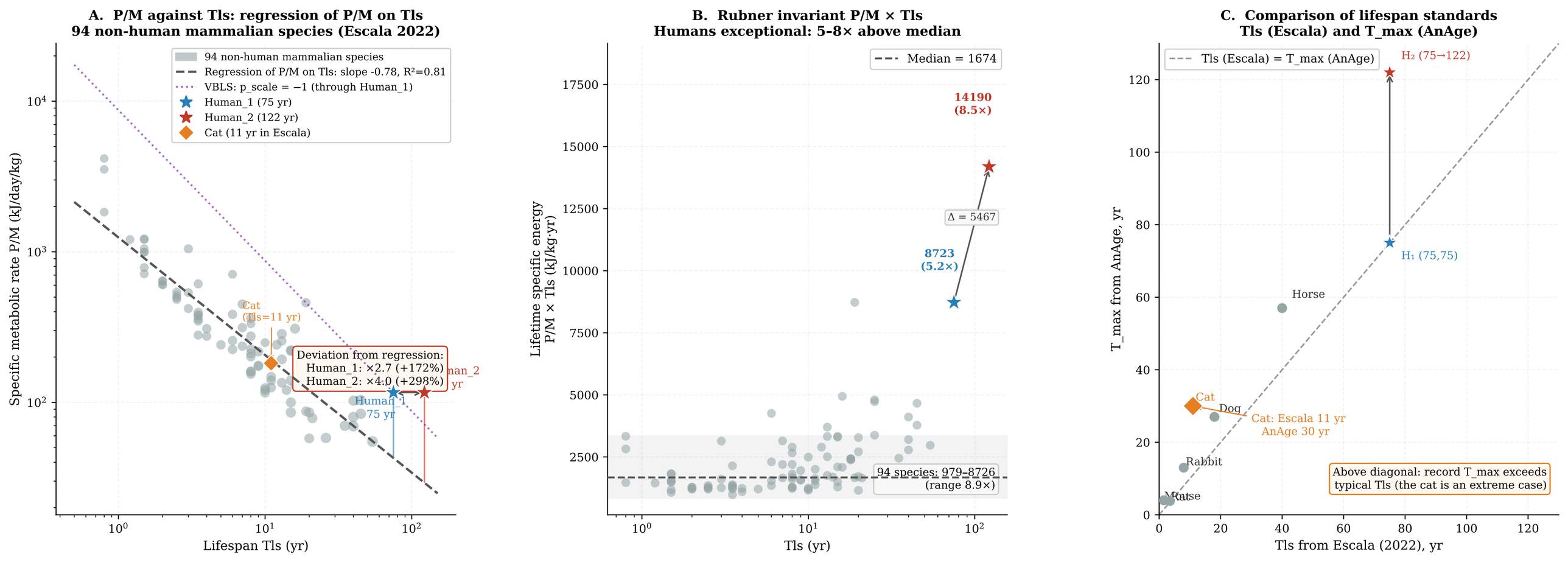

A. P/M against Tls: regression of P/M on Tls
94 non-human mammalian species (Escala 2022)
94 non-human mammalian species
Regression of P/M on Tls: slope -0.78, R²=0.81
VBLS: p_scale = −1 (through Human_1)
Human_1 (75 yr)
Human_2 (122 yr)
Cat (11 yr in Escala)
Specific metabolic rate P/M (kJ/day/kg)
Lifespan Tls (yr)
Cat
(Tls=11 yr)
Deviation from regression:
Human_1: ×2.7 (+172%)
Human_2: ×4.0 (+298%)
Human_1
75 yr
B. Rubner invariant P/M × Tls
Humans exceptional: 5–8× above median
Median = 1674
14190
(8.5×)
Δ = 5467
8723
(5.2×)
94 species: 979–8726
(range 8.9×)
Lifetime specific energy
P/M × Tls (kJ/kg·yr)
Tls (yr)
C. Comparison of lifespan standards
Tls (Escala) and T_max (AnAge)
Tls (Escala) = T_max (AnAge)
H₂ (75→122)
H₁ (75,75)
Horse
Cat
Dog
Cat: Escala 11 yr
AnAge 30 yr
Rabbit
Mouse
Above diagonal: record T_max exceeds
typical Tls (the cat is an extreme case)
T_max from AnAge, yr
Tls from Escala (2022), yr